

**UTM**
UNIVERSITI TEKNOLOGI MALAYSIA

Fakulti Komputeran

FINAL EXAMINATION SEMESTER 2 2013/2014

COURSE CODE : **SCSI 2143**
COURSE NAME : **PROBABILITY AND STATISTICAL DATA ANALYSIS**
YEAR/PROGRAMME :
TIME : **3 HOURS**
DATE :
VENUE :

(ARAHAN KEPADA CALON) :
1. ANSWER ALL QUESTIONS IN THE ANSWER BOOK.
2. ANSWER QUESTION 5(i)(c) ON GRAPH SHEET GIVEN.
3. DO CALCULATIONS IN 2 DECIMAL POINTS.

(Please write your lecturer's name and section in your answer book)

Name	
IC No.	
Year/Programme	
Section	
Lecturer's Name	

QUESTION 1 (10 MARKS)

Mendelian genetics conducted his famous genetics experiments with peas, one sample of offspring consisted of 428 green peas and 152 yellow peas.

- (i) Compute a point estimate of yellow peas. (2 marks)
- (ii) Construct and interpret a 95% confidence interval estimate of the percentage of yellow peas. (5 marks)
- (iii) Based on his theory of genetics, Mendel expected that 25% of the offspring peas would be yellow. Given that the percentage of offspring yellow peas is not 25%, do the results contradict Mendel's theory? Justify your answer. (3 marks)

QUESTION 2 (15 MARKS)

- (i) Company XYZ distributes chocolate drink known as Micho in bottles labelled 32 ml. Company ABC, an auditor company, randomly selects 50 of these bottles, measures their content and obtains a sample mean of 31.8 ml and a sample standard deviation of 0.75 ml. Using a 0.01 significance level, test the auditor company claims that the Company XYZ is cheating consumers. The Company XYZ is cheating consumers if it distributions bottles containing less than 32 ml of Micho.
 - a. State the null and alternative hypotheses. (2 marks)
 - b. Determine and calculate the appropriate test statistic. (2 marks)
 - c. Find the critical value. (2 marks)
 - d. Draw and shade the critical region (rejection area) and state the critical value in the distribution. (2 marks)
 - e. Is it true that the Company XYZ is cheating consumer? (1 mark)

- (ii) In a study of computer use, 25 randomly selected users were asked how much time they spend using computer in a typical week. The time is assumed normally distributed and the sample standard deviation was 5 hours. At $\alpha = 0.05$, is there enough evidence to reject the claim that the population variance is 35 hours?

- State the null and alternative hypotheses. (2 marks)
- Determine and compute the appropriate test statistic. (2 marks)
- Find the critical value. (1 mark)
- What can you conclude from this test? (1 mark)

QUESTION 3 (20 MARKS)

- (i) James and his sister, Alison, each deliver 30 newspapers and each are paid the same amount. Alison claims that the equal payment is unfair as she takes on average longer to distribute the newspaper compare to her brother, James. To test her claim, her father records their delivery times (in minutes) as shown below.

Table 1: Delivery times (minutes)

James	Alison
45	49
30	42
39	39
32	45
34	38
43	49
38	43
35	36
43	33
39	41
32	36
34	-
$\bar{x}_1 = 30$ $s_1 = 4.92$	$\bar{x}_1 = 33$ $s_1 = 5.25$

- a. What is the assumption required for this hypothesis analysis? (1 mark)
- b. State the null and alternative hypotheses. (2 marks)
- c. Variance of both population are unknown and assume both are similar, test whether Alison's claim is justified, using a 5% level of significance. (7 marks)
- d. Based on result in (c), what change would you make to your test procedure so that the conclusion of the hypothesis is different than (c)? (2 marks)

(ii) The following F -test was generated for the data set that contains 37 ceramic strength measurements for two batches of material. The summary statistics for each batch are shown below.

BATCH 1:

NUMBER OF OBSERVATIONS	=	16
MEAN	=	688.9987
STANDARD DEVIATION	=	65.54909

BATCH 2:

NUMBER OF OBSERVATIONS	=	21
MEAN	=	611.1559
STANDARD DEVIATION	=	61.85425

Use 95% significance level to test whether the standard deviation of batch 1 is greater than the standard deviation of batch 2. (8 marks)

QUESTION 4 (25 MARKS)

- (i) A survey was conducted to investigate the age distribution of persons arrested for careless driving. They are distributed in a five different age groups. To test the hypothesis that number of arrested drivers are equal among these five groups, a random sample of 100 persons arrested for careless driving is identified. Given the hypotheses as below,

$$H_0: p_1 = p_2 = p_3 = p_4 = p_5$$

$$H_1: \text{At least one of the } p_i \text{ is not equal}$$

- If the resulting of χ^2 value were 16.5, what conclusion would you reach when using a test with significance level 0.01? (2 marks)
 - What conclusion would be appropriate at significance level 0.05 if the resulting of χ^2 value were 8.2? (2 marks)
 - If there were only four groups of age rather than five groups, what would you conclude if χ^2 value were 9.5 and a test with $\alpha = 0.025$ was used? (2 marks)
- (ii) A presidential candidate hired a public relations firm to conduct a poll to see whether he faces a gender gap with voters. That is, he wants to know whether gender is related to how a voter feels about him as a candidate. The poll asked 800 men and 750 women how they felt about the candidate. The responses are in the table below. Perform the chi-square test to determine whether this politician faces a gender gap.

Table 3

	Strongly support candidate	Strongly oppose candidate	Undecided
Men	325	151	324
Women	258	241	251

- State your hypotheses. (2 marks)
- Find the critical value for this test with $\alpha = 0.005$. (2 marks)
- Construct a table to calculate the observed and expected counts. Then calculate the test statistic. (13 marks)
- Based on answer in (b) and (c), state a conclusion about the results of your test. (2 marks)

QUESTION 5 (30 MARKS)

- (i) Given Table 4 as below where x is the number of hours studied per day by a student, and y is the student's class test marks (assuming that the capability of students are similar).

Table 4

Student Variable	1	2	3	4	5	6	7
x	10	11	9	12	8	7	13
y	80	83	78	86	73	70	89

- Name the two most popular correlation coefficients. Which one will you use for the data presented in the above table and why? (3 marks)
- Calculate the correlation coefficient, r . (8 marks)
- Make a scatter plot of the data. From the plot, does the word linear really provide the most effective description of the relationship between x and y ? (5 marks)
- Based on answers in (b) and (c), what the value of r suggests about the relationship between the numbers of hours studied per day by a particular student and the student's class test marks? (2 marks)

- (ii) Auditors often are required to compare the audited (or current) value of an inventory item with the book (or listed) value. If a company is keeping its inventory and books up to date, there should be a strong linear relationship between the audited and book values. A sample of ten inventory items from a certain company gave the accompanying data on audited and book values.

Table 5

Item	Audit Value, y_i	Book Value, x_i
1	9	10
2	14	12
3	7	9
4	29	27
5	45	47
6	39	42
7	40	36
8	28	21
9	60	59
10	70	67

- a. Fit the model $Y = \beta_0 + \beta_1 x + \varepsilon$ to these data using the method of least squares. (8 marks)
- b. What is your estimate for the expected change in audited value for a one-unit change in book value? (2 marks)
- c. If the book value is $x = 100$, what would you use to estimate the audited value? (2 marks)

List of Formula:

(a) Hypothesis Testing for one sample

$$\text{Test Statistic, } z_0 = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

$$\text{Test Statistic, } T_0 = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

$$\text{Test Statistic, } \chi_0^2 = \frac{(n-1)s^2}{\sigma^2}$$

(b) Hypothesis Testing for two samples

$$\text{Test Statistic, } z_0 = \frac{\bar{x}_1 - \bar{x}_2 - \Delta_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$\text{Test Statistic, } T_0 = \frac{\bar{x}_1 - \bar{x}_2 - \Delta_0}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}; S_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$\text{Test Statistic, } T_0 = \frac{\bar{x}_1 - \bar{x}_2 - \Delta_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}; v = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(s_1^2/n_1\right)^2}{n_1 - 1} + \frac{\left(s_2^2/n_2\right)^2}{n_2 - 1}}$$

$$\text{Test Statistic, } F_0 = \frac{s_1^2}{s_2^2}$$

(c) Chi-Square Test

$$\text{Test Statistic, } \chi_0^2 = \sum_i^n \frac{(o_i - e_i)^2}{e_i}$$

Degree of Freedom for two – way contingency table = $(r - 1)(c - 1)$

(d) Correlation

$$\text{Pearson Correlation Coefficient, } r = \frac{\sum xy - \frac{(\sum x \sum y)}{n}}{\sqrt{\left(\sum x^2 - \frac{(\sum x)^2}{n}\right) \left(\sum y^2 - \frac{(\sum y)^2}{n}\right)}}$$

$$\text{Spearman Correlation Coefficient, } r_s = 1 - \frac{6 \sum d_i^2}{n(n^2-1)} \text{ where } d_i = y_i - x_i$$

$$\text{Test Statistic, } t = \frac{r}{\sqrt{\frac{1-r^2}{n-2}}}$$

(e) Regression

$$b_1 = \frac{\sum xy - \frac{(\sum x \sum y)}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}}$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

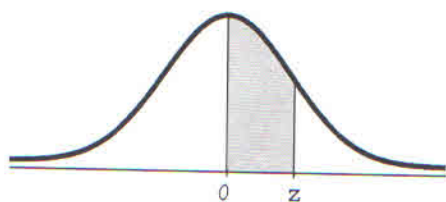


TABLE A-2 Standard Normal (z) Distribution

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.0000	.0040	.0080	.0120	.0160	.0199	.0239	.0279	.0319	.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2257	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2517	.2549
0.7	.2580	.2611	.2642	.2673	.2704	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441
1.6	.4452	.4463	.4474	.4484	.4495	* .4505	.4515	.4525	.4535	.4545
1.7	.4554	.4564	.4573	.4582	.4591	↑ .4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4761	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4864	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936
2.5	.4938	.4940	.4941	.4943	.4945	.4946	.4948	.4949	* .4951	.4952
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962	↑ .4963	.4964
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974
2.8	.4974	.4975	.4976	.4977	.4977	.4978	.4979	.4979	.4980	.4981
2.9	.4981	.4982	.4982	.4983	.4984	.4984	.4985	.4985	.4986	.4986
3.0	.4987	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990
3.10 and higher	.4999									

NOTE: For values of z above 3.09, use 0.4999 for the area.

*Use these common values that result from interpolation:

z score	Area
1.645	0.4500
2.575	0.4950

From Frederick C. Mosteller and Robert E. K. Rourke, *Sturdy Statistics*, 1973, Addison-Wesley Publishing Co., Reading, MA. Reprinted with permission of Frederick Mosteller.

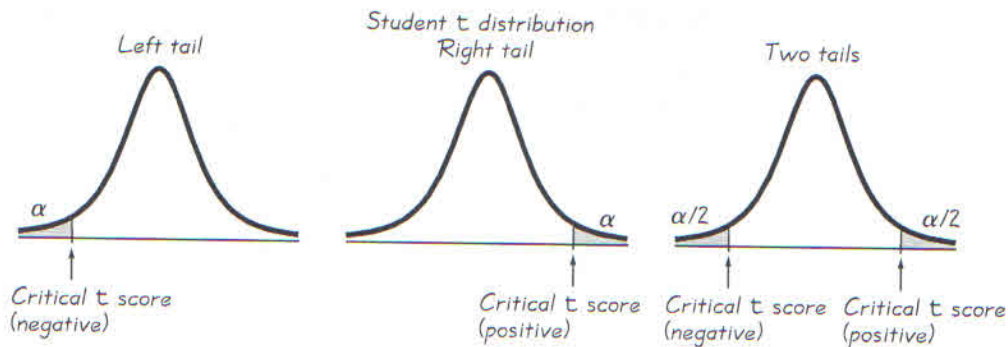


TABLE A-3 t Distribution

Degrees of Freedom	α					
	.005	.01	.025	.05	.10	.25
	(one tail) .01 (two tails)	(one tail) .02 (two tails)	(one tail) .05 (two tails)	(one tail) .10 (two tails)	(one tail) .20 (two tails)	(one tail) .50 (two tails)
1	63.657	31.821	12.706	6.314	3.078	1.000
2	9.925	6.965	4.303	2.920	1.886	.816
3	5.841	4.541	3.182	2.353	1.638	.765
4	4.604	3.747	2.776	2.132	1.533	.741
5	4.032	3.365	2.571	2.015	1.476	.727
6	3.707	3.143	2.447	1.943	1.440	.718
7	3.500	2.998	2.365	1.895	1.415	.711
8	3.355	2.896	2.306	1.860	1.397	.706
9	3.250	2.821	2.262	1.833	1.383	.703
10	3.169	2.764	2.228	1.812	1.372	.700
11	3.106	2.718	2.201	1.796	1.363	.697
12	3.054	2.681	2.179	1.782	1.356	.696
13	3.012	2.650	2.160	1.771	1.350	.694
14	2.977	2.625	2.145	1.761	1.345	.692
15	2.947	2.602	2.132	1.753	1.341	.691
16	2.921	2.584	2.120	1.746	1.337	.690
17	2.898	2.567	2.110	1.740	1.333	.689
18	2.878	2.552	2.101	1.734	1.330	.688
19	2.861	2.540	2.093	1.729	1.328	.688
20	2.845	2.528	2.086	1.725	1.325	.687
21	2.831	2.518	2.080	1.721	1.323	.686
22	2.819	2.508	2.074	1.717	1.321	.686
23	2.807	2.500	2.069	1.714	1.320	.685
24	2.797	2.492	2.064	1.711	1.318	.685
25	2.787	2.485	2.060	1.708	1.316	.684
26	2.779	2.479	2.056	1.706	1.315	.684
27	2.771	2.473	2.052	1.703	1.314	.684
28	2.763	2.467	2.048	1.701	1.313	.683
29	2.756	2.462	2.045	1.699	1.311	.683
Large (z)	2.575	2.326	1.960	1.645	1.282	.675

Formulas and Tables
for *Essentials of Statistics*, by Mario F. Triola
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TABLE A-4 Chi-Square (χ^2) Distribution										
Degrees of Freedom	Area to the Right of the Critical Value									
	0.995	0.99	0.975	0.95	0.90	0.10	0.05	0.025	0.01	0.005
1	—	—	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.071	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.299
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997
21	8.034	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932	41.401
22	8.643	9.542	10.982	12.338	14.042	30.813	33.924	36.781	40.289	42.796
23	9.260	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	18.114	36.741	40.113	43.194	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278	50.993
29	13.121	14.257	16.047	17.708	19.768	39.087	42.557	45.722	49.588	52.336
30	13.787	14.954	16.791	18.493	20.599	40.256	43.773	46.979	50.892	53.672
40	20.707	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691	66.766
50	27.991	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154	79.490
60	35.534	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379	91.952
70	43.275	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425	104.215
80	51.172	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329	116.321
90	59.196	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116	128.299
100	67.328	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807	140.169

From Donald B. Owen, *Handbook of Statistical Tables*. ©1962 Addison-Wesley Publishing Co., Reading, MA. Reprinted with permission of the publisher.

F-table for One-Tailed Test at $\alpha = 0.05$ (95% Confidence Level)

$v_2 \backslash v_1$	1	2	3	4	5	6	7	8	9	10	12	15	20
1	161.4	199.5	215.7	224.6	230.2	234.0	236.8	238.9	240.5	241.9	243.9	245.9	248.0
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45
3	10.13	9.552	9.277	9.117	9.013	8.941	8.887	8.845	8.812	8.786	8.745	8.703	8.660
4	7.709	6.944	6.591	6.388	6.256	6.163	6.094	6.041	5.999	5.964	5.912	5.858	5.803
5	6.608	5.786	5.409	5.192	5.050	4.950	4.876	4.818	4.772	4.735	4.678	4.619	4.558
6	5.987	5.143	4.757	4.534	4.387	4.284	4.207	4.147	4.099	4.060	4.000	3.938	3.874
7	5.591	4.737	4.347	4.120	3.972	3.866	3.787	3.726	3.677	3.637	3.575	3.511	3.445
8	5.318	4.459	4.066	3.838	3.687	3.581	3.500	3.438	3.388	3.347	3.284	3.218	3.150
9	5.117	4.256	3.863	3.633	3.482	3.374	3.293	3.230	3.179	3.137	3.073	3.006	2.936
10	4.965	4.103	3.708	3.478	3.326	3.217	3.135	3.072	3.020	2.978	2.913	2.845	2.774
11	4.844	3.982	3.587	3.357	3.204	3.095	3.012	2.948	2.896	2.854	2.788	2.719	2.646
12	4.747	3.885	3.490	3.259	3.106	2.996	2.913	2.849	2.796	2.753	2.687	2.617	2.544
13	4.667	3.806	3.411	3.179	3.025	2.915	2.832	2.767	2.714	2.671	2.604	2.533	2.459
14	4.600	3.739	3.344	3.112	2.958	2.848	2.764	2.699	2.646	2.602	2.534	2.463	2.388
15	4.534	3.682	3.287	3.056	2.901	2.790	2.707	2.641	2.588	2.544	2.475	2.403	2.328
16	4.494	3.634	3.239	3.007	2.852	2.741	2.657	2.591	2.538	2.494	2.425	2.352	2.276
17	4.451	3.592	3.197	2.965	2.810	2.699	2.614	2.548	2.494	2.450	2.381	2.308	2.230
18	4.414	3.555	3.160	2.928	2.773	2.661	2.577	2.510	2.456	2.412	2.342	2.269	2.191
19	4.381	3.552	3.127	2.895	2.740	2.628	2.544	2.477	2.423	2.378	2.308	2.234	2.155
20	4.351	3.493	3.098	2.866	2.711	2.599	2.514	2.447	2.393	2.348	2.278	2.203	2.124

v_1 = degrees of freedom in numerator

v_2 = degrees of freedom in denominator